

Robotic VR Massage System: Physical Care Robot with Out-of-body Experience in Virtual *Karesansui*

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ABSTRACT

We have developed a robotic VR system that combines a physical care robot with a visual massage therapy in virtual environments. The robotic finger generates a periodic pressing pattern designed using learning data from skilled therapists. The virtual massage movement can be seamlessly synchronized with the movements of the physical care robot controlled in the real world using a high-sensitive force sensor on the robot end-effector. This creates an out-of-body experience in the third-person perspective. Furthermore, the robotic VR massage system provide a healing impression through the design of *Karesansui*, a Japanese dry garden related to *Zen*.

Keywords: Robotic VR, Physical care, Massage therapy, Out-of-body experience.

Index terms: Computing methodologies~Artificial intelligence~Planning and scheduling~Robotic planning~Evolutionary robotics

1 INTRODUCTION

Many people around the world often suffer from musculoskeletal pain, such as stiff shoulders and back pain, and there is an increasing demand for effective massage therapy to relieve this type of pain. The system adapts to individual differences in each recipient's body to provide accurate massage. By merging with extended VR technology, the system can change the social impression of mechanical robots and demonstrate the safety, warmth, and comfort of the massage process [1]. The paper reports on a newly constructed robotic VR massage system that combines a physical care robot with virtual massage therapy.

2 BIBLIOGRAPHY INSTRUCTIONS

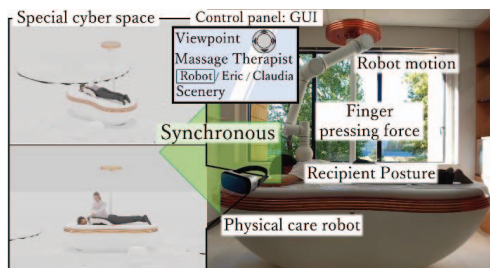


Figure 1: Overview of a robotic VR massage system.

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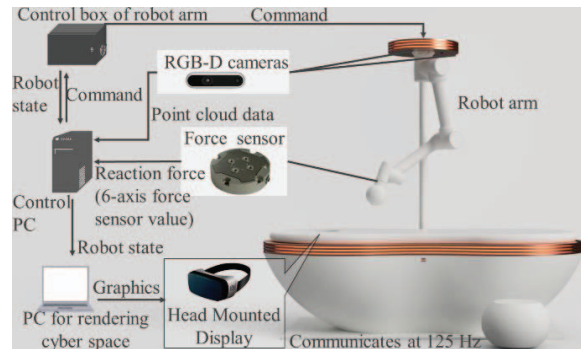


Figure 2: Physical care robotic bed with a sensor fusion-based collaborative robot motion control.

The robotic VR massage system comprises a bed, a multi-axis robot, robot vision and force sensors, and additionally synchronizes with a head-mounted display (HMD) and vital sensors (Figure 1). The sensors detect the robot's motions, the finger pressing force, and the recipient's body state in real-time. The virtual massage and its environment are simultaneously displayed on the HMD.

3 PHYSICAL CARE ROBOT

The autonomous robot shown in Figure 2 adjusts the massage points and the finger pressing force to accommodate individual differences in the human body and physical features. Initially, the depth camera information is used to roughly determine the massage points. The robot control system then approaches the robotic arm and hand to directly touch the human skeletal muscles.

3.1 Human Skeletal Muscle Tracking

The precise location for massage is determined by the skeletal points in the 2D image, as shown in Figure 3(a). The 3D information is then used to estimate the skeletal position. The muscle points are determined based on the skeletal positions. Once the robot touches the muscle point, it detects the surface, stiffness, and size around the target points, as shown in Figure 3(b) [2].

3.2 Finger Pressing Force Tracking and Imitation of Expert Therapy Techniques

Expert massage therapy technique was analyzed by focusing on the behavior of thumb pressing force on a recipient's lower back. The periodic pressing force pattern consists of pushing (red area), keeping constant (yellow area), and pulling back (blue area). The robotic massage adjusts the additional force by integrating a repetitive control algorithm that generates the periodic motion input for human breathing and other movements [3]. Figure 4 shows the time behavior of periodic finger pressing force for each control method.

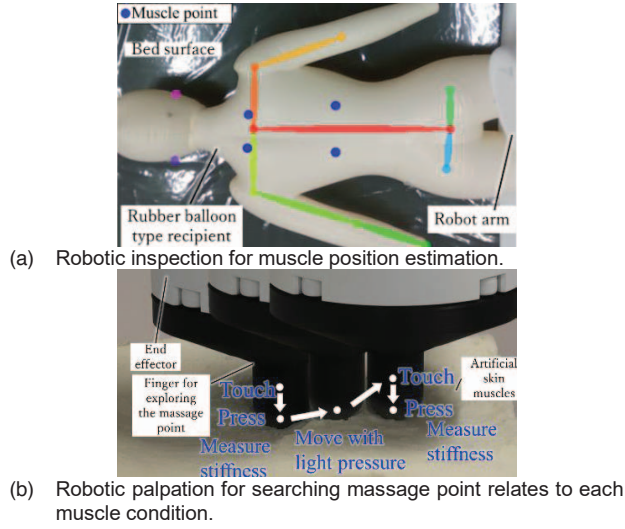


Figure 3: Skeletal muscle tracking system using multi sensors.

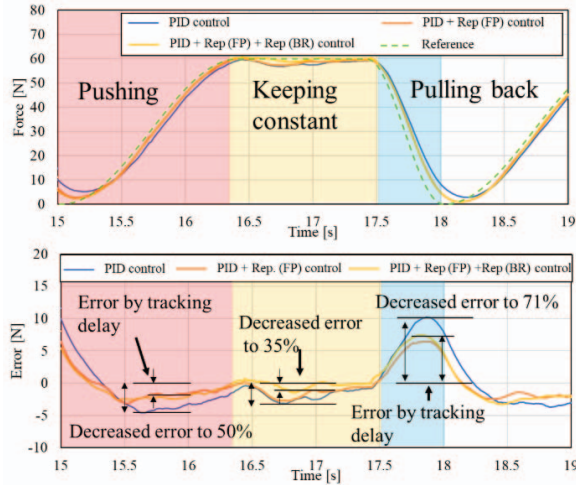


Figure 4: Time behavior of periodic finger pressing force to relieve lower back pain. (Movement in one period).

A large negative error occurred in the pushing section, and a large positive error occurred in the pulling back section. RMSE for the target force is reduced to 71.1 % by Rep(FP) and RMSE in the keeping constant section is reduced to 35.2 % by using Rep(BR).

4 SPATIAL AND TEMPORAL CYBER MASSAGE SPACE

During actual massage therapy, an HMD displayed a cyber space that was synchronized with the physical care robot and the recipient's body state. The cyber massage space was created with a motif of *Karesansui*, which is a Japanese dry garden that expresses natural scenery such as mountains and rivers using only stones. *Karesansui* has a meaning as a space for calming the mind and Zen. The robotic VR massage system has a unique feature of the spatial switching and the therapist changes in the virtual environment. The virtual robot also moves seamlessly in synchronization with the massage motions of the real robot, providing a deep and strong sensation through visual and the physical mutual stimulation. This VR system enables the user to move their viewpoint freely (Figure 5) This allows the user to experience an out-of-body sensation by combining touch and vision [4].

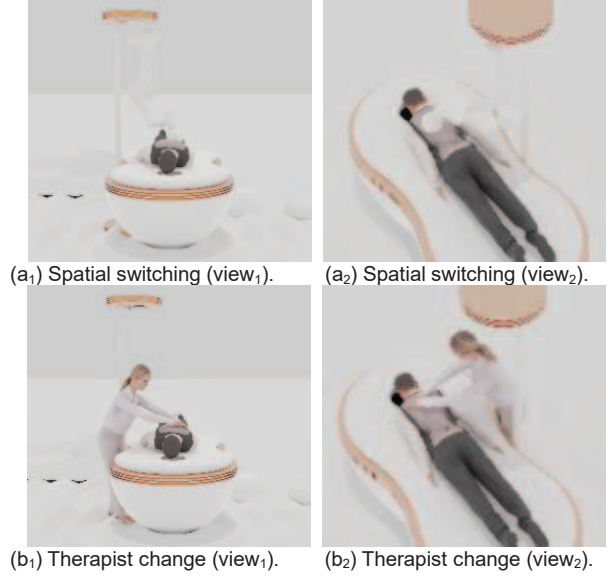


Figure 5: Robotic VR massage experiments by synchronously affecting the visual and the physical senses.

5 CONCLUSION

The robotic VR massage system provides physical and virtual interaction simultaneously to stimulate human senses. It arranges therapy motions based on current body information and allows for changing virtual therapists and massage environment. The system estimates massage points and tracks finger pressing force to account for individual differences in real-time. The massage point can be determined by estimating the position of the target muscle using depth cameras and approaching the muscle's stiff point using force sensing information. Visual information synchronized with the massage is displayed in the real world through HMD.

Multi-sensory synchronous stimulation, such as visual, haptic and voice interaction in the human-robot system is expected to improve the effectiveness of therapy and promote the physical and mental relaxation. Our system provides an experimental platform to investigate the close link between physical and virtual realms and their high level of stimulation of human senses.

ACKNOWLEDGMENTS

This research was supported by JST OPERA JPMJOP1834, JSPS KAKENHI JP22KK0158, JP23H03882, JST SPRING JPMJSP2103

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